



الجامعة الإسلامية العالمية ماليزيا
INTERNATIONAL ISLAMIC UNIVERSITY MALAYSIA
يُونُسُ بَرَسِيَّتِي إِسْلَامُ، إِنْتَارِ بَعْثَا مِلْدِيَّيَا

Garden of Knowledge and Virtue

BICS 1304 - OBJECT ORIENTED PROGRAMMING

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PROJECT:

By GROUP ELATION - SOLAT TRACKER

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1.0 Introduction

The Solat Tracker project is developed to assist IIUM students in maintaining consistency with their daily solat despite demanding academic schedules and personal commitments. Many students struggle with reminders and tracking, which often leads to missed prayers. By applying Object-Oriented Programming (OOP) principles, this project demonstrates how software design can solve real-world problems. The system enables students to track and monitor prayers, get the prayer times, and reminders for any missed prayers in a way that encourages discipline among students.

2.0 Problem Statement

IIUM students often struggle to maintain consistency in performing their daily solat due to busy academic schedules, multiple commitments, and lack of effective reminders. This results in missed prayers and reduced spiritual discipline. A software solution is needed to help students track their prayers, receive timely alerts, and stay motivated through streak-based progress.

3.0 Proposed Solution

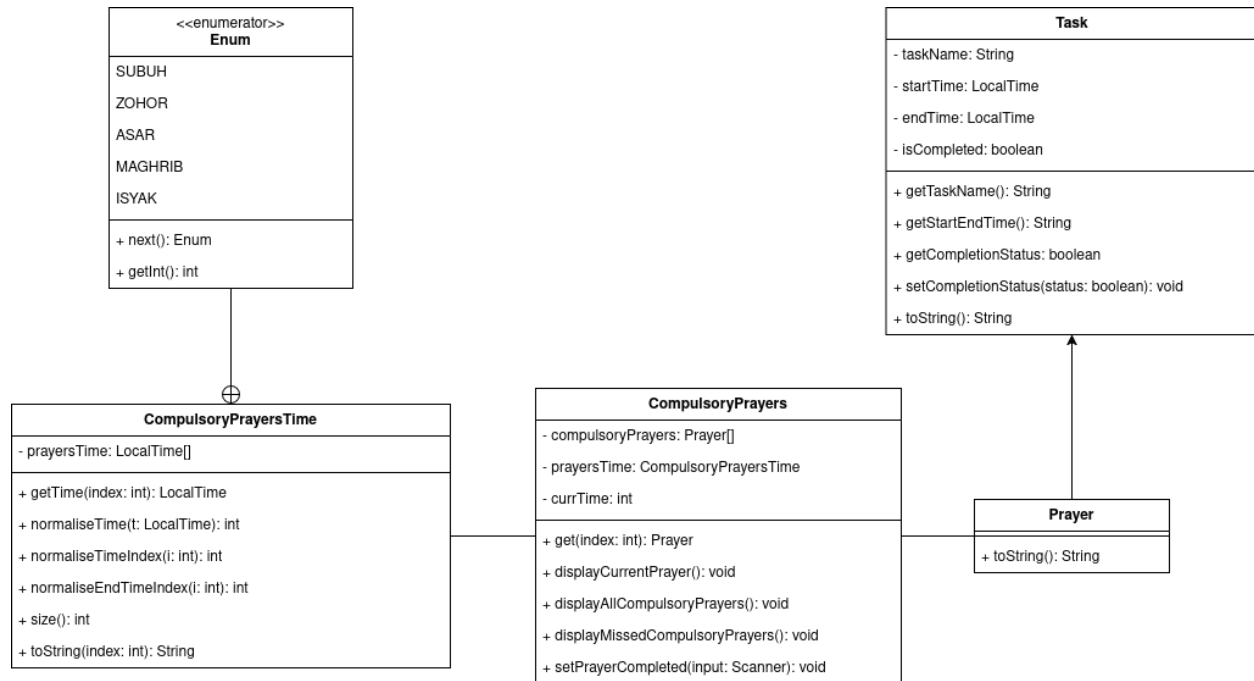
The Solat Tracker is designed to support students in fulfilling their spiritual obligations while balancing academic life. Built using Object-Oriented Programming (OOP) principles, the system allows users to:

- Log completed prayers.
- Get prayer times for the current day.
- See current prayer.
- See missed prayers.
-

Main Classes

- **CompulsoryPrayers** – represents the five obligatory daily prayers.
- **CompulsoryPrayersTime** – defines fixed times for compulsory prayers and an enum.
- **Prayer** – child class of Task for prayers.
- **Task** – represents tasks or reminders linked to prayers.
- **Main** – entry point that runs and demonstrates the system.

4.0 UML Diagram



5.0 Sample Run

Screenshots of the main page, reminder setup, streak tracking, and notification flow will be included to demonstrate system functionality.

5.1 Main Page

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: |
```

5.2 Compulsory Prayer Time

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: 1

SUBUH prayer
Prayer time: 06:00 - 13:19
Prayer completion: No

ZOHOR prayer
Prayer time: 13:20 - 16:39
Prayer completion: No

ASAR prayer
Prayer time: 16:40 - 19:29
Prayer completion: No

MAGHRIB prayer
Prayer time: 19:30 - 20:29
Prayer completion: No

ISYAK prayer
Prayer time: 20:30 - 05:59
Prayer completion: No
```

5.3 Display Current Prayer

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: 2

ISYAK prayer
Prayer time: 20:30 - 05:59
Prayer completion: No
```

5.4 Missed Prayer(s)

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: 3

SUBUH prayer
Prayer time: 06:00 - 13:19
Prayer completion: No

ZOHOR prayer
Prayer time: 13:20 - 16:39
Prayer completion: No

ASAR prayer
Prayer time: 16:40 - 19:29
Prayer completion: No

MAGHRIB prayer
Prayer time: 19:30 - 20:29
Prayer completion: No
```

5.5 Set Prayer as Completed

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: 4

Prayer List
SUBUH
ZOHOR
ASAR
MAGHRIB
ISYAK

From the above list, enter which prayer you have completed:
zohor
```

5.6 Exit Program

```
Choices Menu
1. Display Compulsory Prayers Time
2. Display Current Prayer
3. Display Missed Prayers
4. Set A Prayer As Completed
5. Exit
Enter choice: 5
Thank you for using the program! Exiting...
ahsan@Santop:~/Documents/BCS UIAM/BICS 1304 OOP/Exercise/Coding$ |
```

6.0 Conclusion

The Solat Tracker integrates Islamic values with modern technology, helping students balance spiritual discipline and academic life. By applying OOP principles, the project demonstrates how software design can solve real-world problems while fostering consistency, motivation, and community spirit among IIUM students.